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A bundle of connectivity

RS9113 chipset comes with the following features that help the user design a machine to machine (M2M) device easily: Wi-Fi 2.4 GHz, Wi-Fi 5.8 GHz, Bluetooth 2.1+EDR (enhanced data rate), BT 3.0, BT 4.0 (also known as Bluetooth Smart, Bluetooth Low Energy or Bluetooth LE), ZigBee, calculation power on internal MCU for your application, a large number of diverse peripheral interfaces like I²C (Inter-Integrated Circuit), I²S (Integrated Interchip Sound), SPI (Serial Peripheral Interface), QSPI (Quad Serial Peripheral Interface), UART (Universal Asynchronous Receiver/Transmitter), GPIO (General-Purpose Input/Output), JTAG (Joint Test Action Group) and analogue input/output.

with small host controllers. With the multi-threaded processor and on-chip ROM and RAM, this chipset enables integration into zero host-load applications with the capability to run the Wi-Fi security supplicant and TCP/IP on the chip itself.

What's special about it?

The device can simultaneously maintain connection on some or all of the interfaces, providing multi-protocol gateway for users. It has an antenna and a U.FL connector (a miniature RF connector) for external antenna integrated on it with an option to select either one of them. What's also interesting is that all wireless transceivers inside the chipset will end in a single antenna.

The innovative technology implemented by Redpine will ensure that Wi-Fi, classic Bluetooth, Bluetooth LE and ZigBee on 2400 MHz will not interfere. This new coexistent algorithm helps system design engineers to solve the problems often encountered when trying to co-locate multiple wireless protocols in current multi-chip solutions.

For example, an RS9113 chipset-based gateway device could communicate with a medical sensor via Bluetooth, smartphone via Bluetooth or Wi-Fi, or in-house display device via ZigBee functionality, without the need for using multiple connectivity modules from various vendors.

Redpine offers this product to customers in the form of chips or as ready-to-use wireless modules that can be easily integrated into embedded system designs. They provide a complete reference design and development environment for creating applications using the new chip, and offer an easy-to-use development kit SDIO, SPI,

UART, and USB2.0 interfaces.

The RS9113 solution comes with Redpine's OneBox embedded software framework which supports common API-based base transceiver station, access-point, dual-mode BT 4.0, Wi-Fi Direct, ZigBee and SEP2.0 (Smart Energy Profile 2.0) functions on diverse host platforms and embedded operating systems. Software package includes firmware, reference drivers, application profiles and configuration GUI (graphic user interface) for Windows, Linux and Android operating systems.

This chipset can be used to enable wireless connectivity to a variety of electronic systems like healthcare, industrial and home automation, and in several other devices like VoWiFi (Voice over Wi-Fi) phones, serial to Wi-Fi interface, wireless sensors, smart meters, in-home devices like thermostats and displays, wireless RTLS (real-time locating system) tags and other mobile and M2M communication implementations.

Challenge of developing this SoC

Engineering a complex SoC like this requires careful consideration of factors like meeting power consumption goals, data throughput, device size factor, and performance. These were a major challenge while developing this product, says N Venkatesh of Redpine Signals, Inc. RF and power amplifier design in CMOS is complex. Developing a protocol compliant firmware that is able to run within limited embedded resources was also another challenge faced by the team. It took approximately two years to get this product idea from inception to fruition. ●

The author is a technical correspondent at Efy